ROS 2 TurtleBot3 Simulation Assignment

Progress Report

1. Project Overview

This project involves setting up ROS 2 Jazzy and exploring the TurtleBot3 simulation using Gazebo and RViz2.

2. Environment

- Operating System: Ubuntu 24.04 (WSL2)
- ROS 2 Distribution: Jazzy
- Robot model: TurtleBot3 Burger
- Simulation: Gazebo
- Visualization: RViz2

3. Setup Commands

```
source /opt/ros/jazzy/setup.bash
export TURTLEBOT3_MODEL=burger
```

4. Simulation Launch

Command attempted:

```
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

5. Navigation and Visualization

Navigation launch command attempted:

```
ros2 launch turtlebot3_navigation2 navigation2.launch.py map:=/opt/ros/jazzy/share/turtlebot3_navigation2/map/
```

RViz2 configuration command:

```
ros2 run rviz2 rviz2 -d /opt/ros/jazzy/share/turtlebot3_navigation2/rviz/tb3_navigation2.rviz
```

6. Progress and Challenges

- ROS 2 Jazzy environment was set up.
- TurtleBot3 and Gazebo packages were installed.
- Gazebo and RViz2 were opened and explored.
- Terminal commands were executed to launch the simulation and navigation components.
- Issues remain with displaying the robot, map, and transforms correctly in RViz2.
- Further troubleshooting is required to achieve the expected final output.

7. Evidence

Screenshots and a screen recording have been saved to document the work completed so far.

8. Current Status

The assignment is partially completed. Remaining tasks include resolving the simulation and visualization issues, verifying navigation, and capturing successful final output if possible. This report documents the current progress and does not claim that navigation is working.